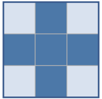
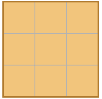
(a) Structuring neighbourhoods

$\mathcal{E}$: min over the 6-neighbour cross



$3\times1\times1$
$1\times3\times1$ merged by min
$1\times1\times3$

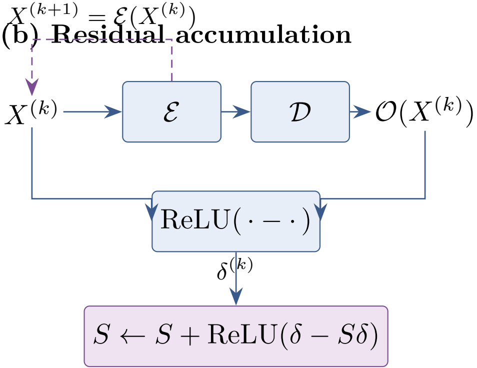
$\mathcal{D}$: max over the full 3^3 cube



all 26 neighbours and the centre

The two neighbourhoods differ, so $\mathcal{O} = \mathcal{D} \circ \mathcal{E}$ is the *implemented* soft opening rather than an adjoint morphological pair.

(b) Residual accumulation



The residual is taken at erosion depth $k = 0$ and after each of the 10 updates, so eleven ridge maps are combined. For $S, \delta \in [0, 1]$ the update is the probabilistic union $S + \delta(1 - S)$.